Notes: Fernández-Villaverde, Hurtado & Nuño (Econometrica, 2023)

Financial Frictions and the Wealth Distribution

Contents

1 Big picture

- **Question.** How can financial frictions and household heterogeneity generate nonlinear aggregate dynamics, time-varying risk, and regime-like movements in leverage, debt, output, and interest rates?
- **Core model.** The paper builds a continuous-time heterogeneous-agent DSGE model with a representative firm, one representative financial expert, and a continuum of households with idiosyncratic labor income risk.
- **Two financial frictions.** First, only the expert can hold capital and the expert cannot issue state-contingent claims. Second, households cannot hold capital and can save only in noncontingent risk-free bonds, subject to a no-shorting constraint.
- **Central mechanism.** Experts supply bonds to lever their capital holdings. Households demand bonds for precautionary saving. Their interaction creates endogenous aggregate risk even though the only exogenous aggregate shock is a capital growth shock.
- **Main nonlinear result.** The model has a unique deterministic steady state, but the nonlinear stochastic economy has multiple stochastic steady states (SSSs): a high-leverage SSS and a low-leverage SSS, plus an unstable middle point.
- **State dependence.** The same negative capital shock is amplified and propagated more when the economy starts in the high-leverage region than when it starts in the low-leverage region.
- **Debt supercycles.** The economy can spend long spells in a high- or low-leverage basin of attraction. These spells resemble debt supercycles: long periods of leverage accumulation and deleveraging.
- **Role of the wealth distribution.** Household heterogeneity is not a detail. It determines precautionary bond demand, which pins down the risk-free rate and justifies the expert's leverage in each regime.
- **Computational contribution.** The paper generalizes the Krusell-Smith algorithm by using a neural network to approximate a fully nonlinear perceived law of motion (PLM) for aggregate debt.
- **Estimation contribution.** After solving the model, the authors show how to evaluate a likelihood function for aggregate and micro observations using inference with diffusions.
- **Economic takeaway.** Financial frictions and wealth heterogeneity can generate endogenous regime switching and large state-dependent impulse responses, even in a model with a unique deterministic steady state.

2 Data, calibration, and measurement

2.1 What is measured

The paper is primarily quantitative and computational rather than reduced-form empirical. The core objects are simulated model variables:

- aggregate capital, K_t;
- aggregate debt, B_t;
- expert net worth or equity, N_t;
- leverage, K_t/N_t;
- output, Y_t;
- wages, w_t;
- the risk-free rate, r_t;
- the return to capital, r_t^k;
- the cross-sectional distribution of household assets and employment states.

Interpretation: The key state variables are not only aggregate capital or productivity. Debt, expert equity, and the household wealth distribution are essential because they determine risk sharing and precautionary saving.

2.2 Household heterogeneity

- Households differ by assets a and idiosyncratic income state z.
- There are two idiosyncratic states, interpreted as unemployment and employment.
- Households receive labor income, consume, and save in risk-free bonds.
- Households cannot short bonds: $a_t \geq 0$.
- The cross-sectional distribution over assets and income states is denoted by G_t.

Interpretation: Household heterogeneity creates a precautionary demand for bonds. This demand is what allows the expert to issue debt and lever capital holdings.

2.3 Baseline parameterization

- Capital share: $\alpha = 0.35$.
- Capital depreciation: $\delta = 0.1$.
- Household risk aversion: $\gamma = 2$.
- Household discount rate: $\rho = 0.05$.
- Transition rates imply a 5 percent unemployment rate and a monthly job-finding rate of 0.3.
- Income states are normalized to keep average labor income equal to one.
- The expert's discount rate is set so that the deterministic steady-state leverage ratio is approximately $K/N = 2$.

- The aggregate capital-shock volatility σ is estimated by maximum likelihood using U.S. quarterly data from 1984Q1 to 2017Q4. The estimate is $\sigma = 0.0142$.

Interpretation: The calibration is transparent: most parameters are standard or tied to labor market targets, while the key aggregate-risk parameter is estimated from U.S. aggregate data.

2.4 High- and low-leverage regions

The model's quantitative solution produces two stable stochastic steady states:

- **High-leverage SSS:** $B^{HL} = 1.9641$, $N^{HL} = 1.7470$, $K^{HL} = 3.7111$, and $K^{HL}/N^{HL} = 2.1243$.
- **Low-leverage SSS:** $B^{LL} = 1.0967$, $N^{LL} = 2.6010$, $K^{LL} = 3.6977$, and $K^{LL}/N^{LL} = 1.4216$.
- The deterministic steady state lies near the high-leverage SSS in terms of capital, but the stochastic economy treats the high- and low-leverage regions very differently.

Interpretation: The regime distinction is not mostly about capital. Capital is similar across the two stable SSSs. The key difference is the allocation of total wealth between experts and households, which changes leverage and risk exposure.

3 Model and solution method

3.1 Firm technology

A representative firm rents aggregate capital K_t and labor L_t to produce

$$Y_t = K_t^\alpha L_t^{1-\alpha}. \tag{1}$$

Competitive factor prices are

$$w_t = (1-\alpha)\frac{Y_t}{L_t}, \qquad r_t^c = \alpha\frac{Y_t}{K_t}. \tag{2}$$

Capital evolves according to

$$\frac{dK_t}{K_t} = (\iota_t - \delta)dt + \sigma dZ_t. \tag{3}$$

Interpretation: The capital shock directly affects the expert because only the expert holds capital. This makes financial amplification transparent.

3.2 Expert balance sheet and net worth

The expert holds capital $\widehat{K}_t$ and finances it with risk-free debt $\widehat{B}_t$. Net worth is

$$\widehat{N}_t = \widehat{K}_t - \widehat{B}_t. \tag{4}$$

Leverage is

$$\widehat{\omega}_t = \frac{\widehat{K}_t}{\widehat{N}_t}. \tag{5}$$

The expert's net worth evolves as

$$d\widehat{N}_t = \left[r_t + \widehat{\omega}_t(r_t^c - \delta - r_t)\right]\widehat{N}_t dt - \widehat{C}_t dt + \sigma\widehat{\omega}_t\widehat{N}_t dZ_t. \tag{6}$$

Interpretation: The expert bears all capital risk. Higher leverage raises expected returns but also raises the sensitivity of net worth to aggregate shocks.

3.3 Household problem

Households have CRRA utility and save in risk-free bonds. Let $c_i(a, G, N)$ denote consumption of a household in idiosyncratic state i. The saving rule is

$$s_i(a, G, N) = wz_i + ra - c_i(a, G, N), \tag{7}$$

with the no-shorting constraint $a \geq 0$.

The full household HJB depends on the infinite-dimensional distribution G. This is difficult because the value function responds to the full cross-sectional distribution.

Interpretation: Household demand for bonds depends on wages, interest rates, idiosyncratic risk, and expectations about aggregate dynamics.

3.4 Perceived law of motion

Following the Krusell-Smith logic, the paper replaces the full distribution G_t with its first moment, aggregate debt B_t. The perceived law of motion is

$$dB_t = h(B_t, N_t)dt. \tag{PLM}$$

Given this PLM, the household HJB becomes a finite-dimensional equation:

$$\rho V_i(a,B,N) = \max_c \Bigg\{ \frac{c^{1-\gamma}-1}{1-\gamma} + s_i(a,B,N)\frac{\partial V_i}{\partial a} + \lambda_i\left[V_j(a,B,N) - V_i(a,B,N)\right] \tag{8}$$

$$+ h(B,N)\frac{\partial V_i}{\partial B} + \mu^N(B,N)\frac{\partial V_i}{\partial N} + \frac{(\sigma^N(B,N))^2}{2}\frac{\partial^2 V_i}{\partial N^2}\Bigg\}. \tag{9}$$

Interpretation: The state space collapses from an infinite-dimensional distribution to two aggregate state variables, B and N, while keeping the law of motion fully nonlinear.

3.5 Neural network approximation

The algorithm is:

1. Guess an initial PLM $h_0(B,N)$.
2. Solve the household HJB given the current PLM.
3. Simulate the model and recover realized changes in aggregate debt.
4. Train a neural network to update the PLM.
5. Iterate until the PLM forecast error is sufficiently small.

Interpretation: The neural network is used because the PLM is strongly nonlinear in aggregate states. A log-linear Krusell-Smith law of motion misses the regime structure.

3.6 Stochastic steady states

A stochastic steady state is a point where debt and expert equity do not drift when the aggregate shock realization is zero. Thus SSSs solve

$$h(B,N) = 0, \qquad \mu^N(B,N) = 0. \tag{10}$$

Interpretation: The paper's main nonlinear result comes from multiple intersections of these two zero-drift curves. The deterministic steady state is unique, but the stochastic economy has multiple SSSs because agents price risk differently across regions.

4 Quantitative design and credibility

4.1 Why the solution is credible

- The paper solves a heterogeneous-agent model with aggregate shocks in continuous time.
- The neural network PLM is evaluated by forecast errors from long simulations.
- Figure 4 shows that the neural-network PLM has far tighter one-month forecast errors than a linear PLM.
- The PLM only uses aggregate debt as a summary of the household distribution. This is reasonable because the household consumption rule is approximately linear in household assets except near the borrowing constraint.
- The paper still warns that if many households were clustered near the borrowing constraint, more moments of the wealth distribution might be needed.

Interpretation: This is a bounded-rationality equilibrium in the Krusell-Smith sense: households forecast with a low-dimensional law of motion, and that law of motion is self-justifying because forecast errors are small.

4.2 Why the model produces endogenous aggregate risk

- In the high-leverage basin, a negative capital shock reduces expert equity sharply.
- Expert debt adjusts slowly because debt accumulation is tied to the slow rebuilding of net worth.
- Capital and output recover more slowly after shocks in the high-leverage region.
- Households understand this higher risk and demand more precautionary bonds.
- Higher precautionary saving lowers the risk-free rate and helps sustain expert leverage.

Interpretation: The model has a feedback loop: leverage raises fragility, fragility raises household precautionary saving, precautionary saving lowers the risk-free rate, and the low risk-free rate makes leverage privately attractive.

4.3 What the paper cannot fully prove

- The paper computes a self-justified equilibrium under a low-dimensional PLM, not the full rational-expectations equilibrium with the entire distribution as a state variable.
- The model abstracts from many real financial institutions, including banks with deposits, default, equity issuance, and nominal rigidities.
- The only aggregate shock is a capital growth shock, which stands in for broader aggregate shocks.
- The model uses calibration and likelihood estimation, not a direct causal empirical design.

Interpretation: The paper's contribution is to show a mechanism and a computational method, not to claim that this is a complete empirical model of U.S. financial cycles.

5 Tables and figures

This section keeps the same *Read* plus *Economic message* format and groups adjacent visuals to save space and improve reading speed. All visuals are directly cropped from the original paper.

5.1 Table I + Figure 1: calibration and deterministic steady state

Parameter	Value	Description	Source/Target
α	0.35	capital share	standard
δ	0.1	capital depreciation	standard
γ	2	risk aversion	standard
ρ	0.05	households' discount rate	standard
λ_1	0.986	transition rate unemp.-to-employment	monthly job finding rate of 0.3
λ_2	0.052	transition rate employment-to-unemp.	unemployment rate 5%
z_1	0.72	income in unemployment state	Hall and Milgrom (2008)
z_2	1.015	income in employment state	$\mathbb{E}(y) = 1$
$\widehat{\rho}$	0.0497	experts' discount rate	$K/N = 2$

Table I: baseline parameterization.

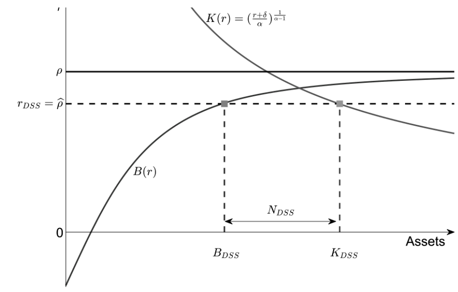


FIGURE 1.—DSS of the model.

Figure 1: deterministic steady state.

Read:

- Table I sets $\alpha = 0.35$, $\delta = 0.1$, $\gamma = 2$, and $\rho = 0.05$.
- The income states and transition rates are calibrated to a 5 percent unemployment rate and a monthly job-finding rate of 0.3.
- The expert's discount rate is set so that deterministic steady-state leverage is approximately $K/N = 2$.
- Figure 1 shows the crossing of capital demand and household bond demand that pins down the deterministic steady state.

Economic message: The deterministic steady state is a useful benchmark, but the stochastic economy behaves very differently because agents price aggregate risk.

5.2 Figures 2–3: nonlinear PLM for debt and equity

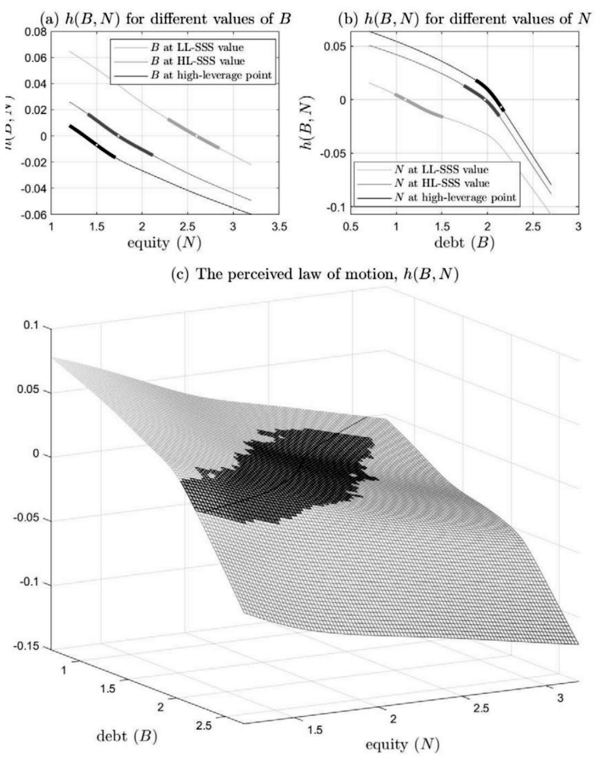


FIGURE 2.—The PLM $h(B,N)$ and transversal cuts.

Figure 2: PLM for aggregate debt $h(B,N)$.

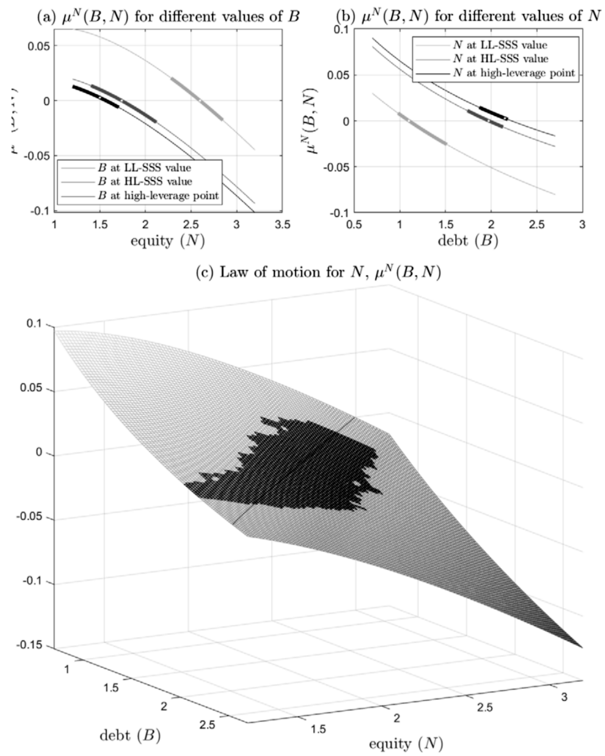


FIGURE 3.—The law of motion $\mu^N(B,N)$ and transversal cuts.

Figure 3: law of motion for equity $\mu^N(B,N)$.

Read:

- Figure 2 plots the perceived law of motion for debt, $h(B,N)$.
- Figure 3 plots the law of motion for expert equity, $\mu^N(B,N)$.
- Both surfaces are nonlinear in debt and equity.
- The thick regions indicate where the ergodic distribution puts probability mass.
- The cuts show different behavior around the high-leverage and low-leverage regions.

Economic message: The nonlinear PLM is the engine of the regime structure. A log-linear rule would smooth away the multiple zero-drift intersections.

5.3 Figures 4–5: forecast accuracy and phase diagram

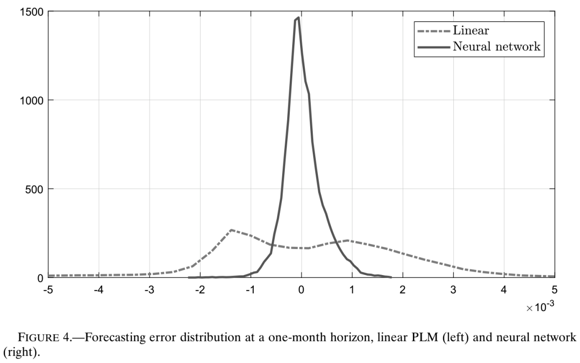


Figure 4: one-month forecast error distribution.

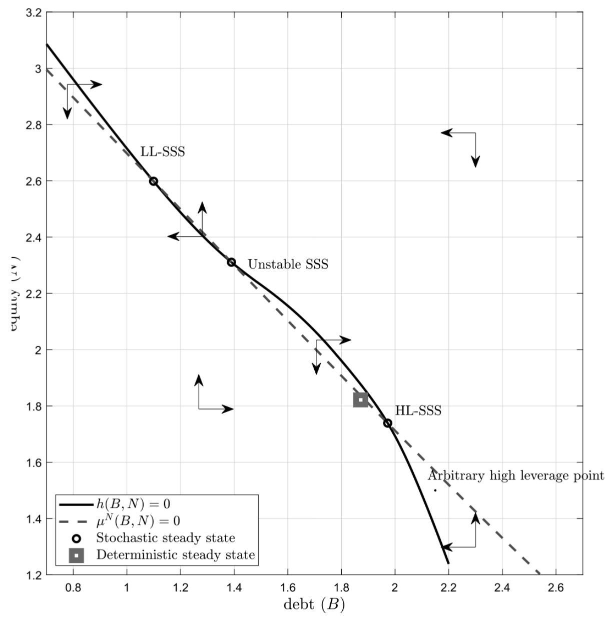


Figure 5: phase diagram, DSS, and SSSs.

Read:

- Figure 4 shows that the neural-network PLM has a tight forecast-error distribution around zero.
- The linear PLM has a much wider forecast-error distribution.
- Figure 5 plots the zero-drift loci $h(B,N) = 0$ and $\mu^N(B,N) = 0$.
- The two loci intersect three times: a stable high-leverage SSS, an unstable middle SSS, and a stable low-leverage SSS.
- The arrows show the direction of adjustment in the debt-equity state space.

Economic message: The economy has a unique deterministic steady state but multiple stochastic steady states. The distinction comes from endogenous risk, not from multiple equilibria.

5.4 Figure 6 + Table II: state-dependent impulse responses and basin moments

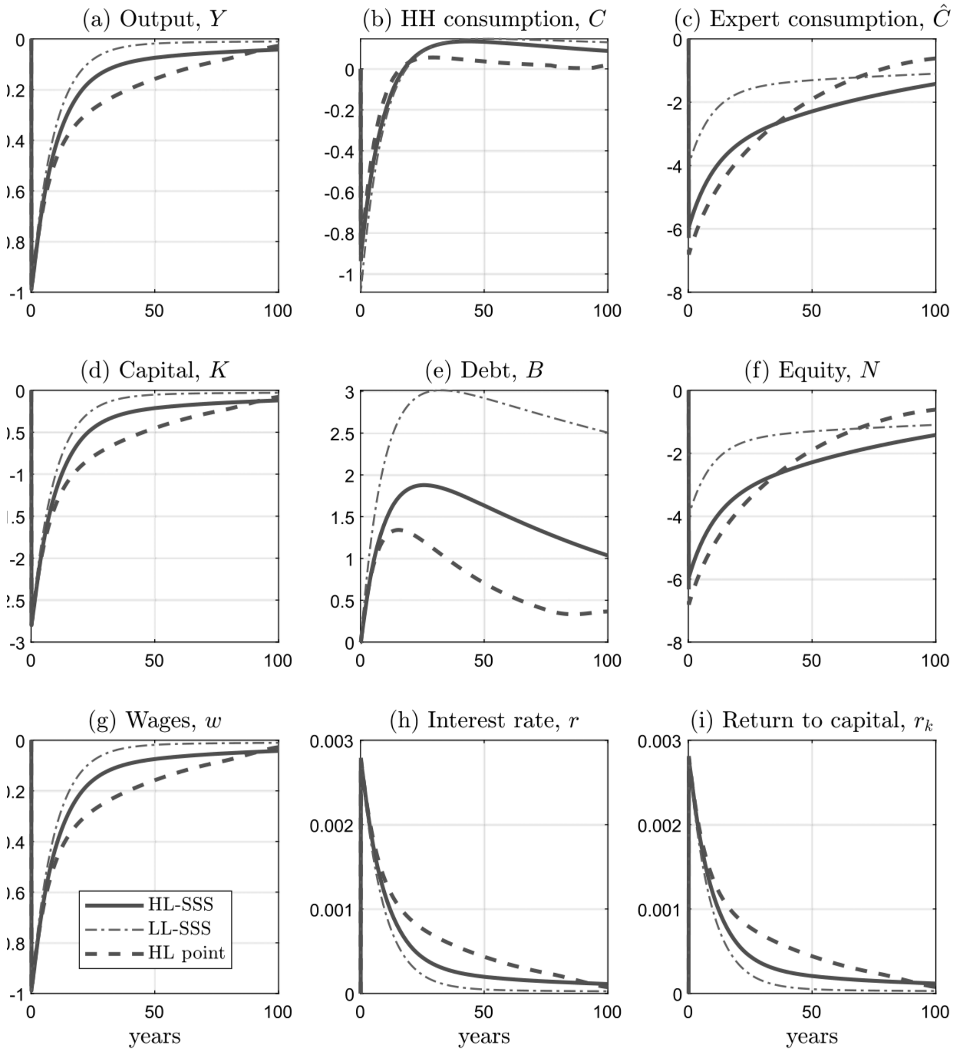


FIGURE 6.—GIRFs for different initial states.

Figure 6: GIRFs for different initial states.

TABLE II

MOMENTS CONDITIONAL ON BASIN OF ATTRACTION.

	Mean	Standard deviation	Skewness	Kurtosis
$Y^{\text{basin HL}}$	1.5802	0.0193	0.0014	2.869
$Y^{\text{basin LL}}$	1.5829	0.0169	0.1186	3.0302
$r^{\text{basin HL}}$	4.92	0.3364	0.0890	2.866
$r^{\text{basin LL}}$	4.89	0.2947	−0.0282	3.0056
$w^{\text{basin HL}}$	1.0271	0.0125	0.0014	2.8691
$w^{\text{basin LL}}$	1.0289	0.0111	0.1186	3.0302

Table II: moments conditional on basin of attraction.

Read:

- Figure 6 compares GIRFs to the same two-standard-deviation negative capital shock.
- Starting from high leverage makes output, capital, debt, and equity responses more persistent.
- Debt is hump-shaped: after the negative shock, debt first rises and then only gradually falls.
- Table II shows that the high-leverage basin has higher output volatility: 0.0193 versus 0.0169.
- Output skewness is near zero in the high-leverage basin but 0.1186 in the low-leverage basin.

Economic message: State dependence is not a small quantitative detail. The same shock has different persistence and risk implications depending on where the economy starts.

5.5 Figures 7–8: equilibrium paths and debt supercycles

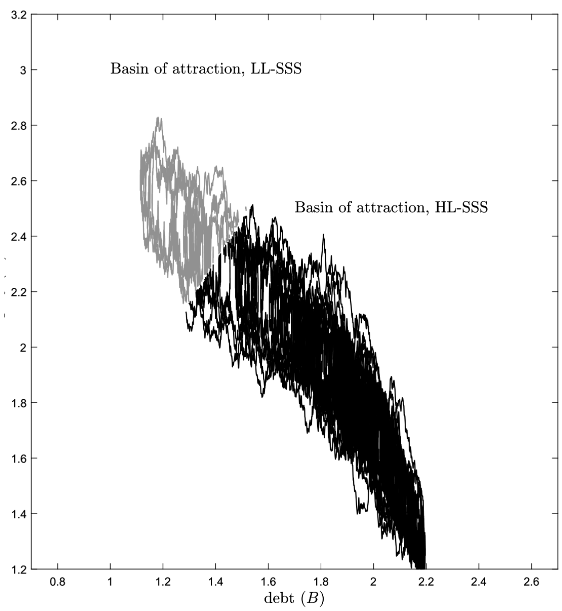


FIGURE 7.—Simulation of equilibrium paths.

Figure 7: simulated equilibrium paths.

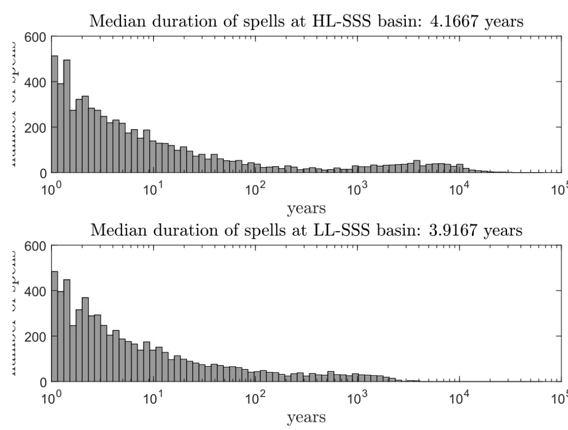


FIGURE 8.—Spell durations at each SSS.

Figure 8: spell durations at each SSS.

Read:

- Figure 7 shows simulated paths in the debt-equity state space.
- Light gray paths lie in the basin of the low-leverage SSS; darker paths lie in the basin of the high-leverage SSS.
- Figure 8 shows that spells can be very long despite a median duration of around four years.
- The distribution has a fat right tail, consistent with persistent borrowing and deleveraging episodes.
- The economy spends about 86.7 percent of the time in the high-leverage basin.

Economic message: The model can generate long-lasting debt supercycles without exogenous regime switching. The regimes are endogenous states of the economy.

5.6 Figures 9–10: ergodic distribution and household wealth distribution

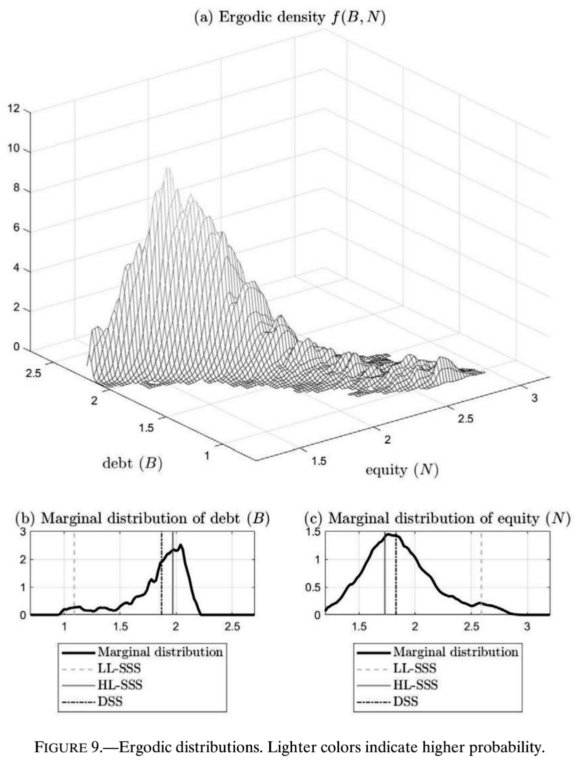


Figure 9: ergodic distribution of debt and equity.

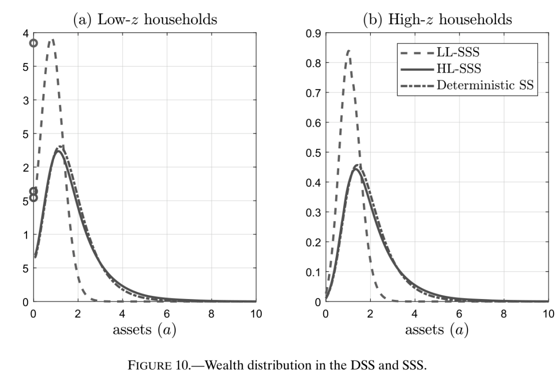


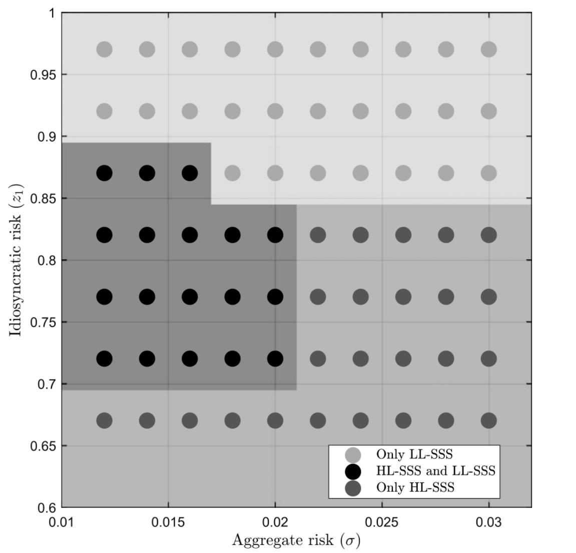
Figure 10: wealth distribution in DSS and SSSs.

Read:

- Figure 9 shows that most probability mass is around high-debt, low-equity states, but there is a tail toward high equity and low debt.
- The marginal debt distribution is more concentrated than the marginal equity distribution.
- Figure 10 compares wealth distributions for low- and high-income households.
- Moving from DSS to high leverage thickens the right tail, while moving toward low leverage shifts the distribution left.
- The wealth Gini is 0.28977 in the DSS, 0.24318 in the LL-SSS, and 0.31935 in the HL-SSS.

Economic message: Financial regimes change the distribution of household wealth. Higher leverage is associated with a thicker right tail and more inequality.

5.7 Figures 11–12: heterogeneity, risk, and decision rules



11.—Multiplicity of SSS(s) as a function of aggregate and idiosyncratic volatili

Figure 11: multiplicity of SSSs by aggregate and idiosyncratic risk.

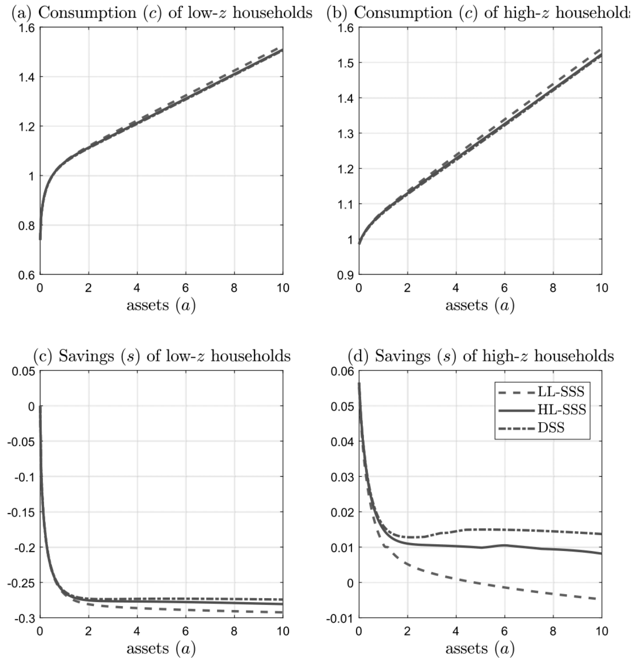


FIGURE 12.—Consumption and saving functions.

Figure 12: consumption and saving functions.

Read:

- Figure 11 varies aggregate volatility and idiosyncratic risk.
- With high idiosyncratic risk, only the high-leverage SSS tends to exist because households save aggressively.
- With low idiosyncratic risk, only the low-leverage SSS tends to exist because precautionary saving is weak.
- In the intermediate region, both stable SSSs coexist.
- Figure 12 shows that consumption and saving rules are close to linear for most asset levels but differ sharply near the borrowing constraint and across aggregate states.

Economic message: Household heterogeneity is the key. Idiosyncratic risk creates bond demand, and bond demand changes the financial sector's leverage and the aggregate regime.

6 Short summary

- The paper studies a continuous-time heterogeneous-agent model with a financial expert and households.
- Only the expert can hold capital; households can save only in risk-free bonds.
- The expert cannot issue state-contingent claims, so expert net worth absorbs capital shocks.
- Households face idiosyncratic labor income risk and demand bonds for precautionary saving.
- The interaction between expert bond supply and household bond demand creates endogenous aggregate risk.
- The model has a unique deterministic steady state but multiple stochastic steady states.
- The stable stochastic steady states correspond to high-leverage and low-leverage regions.
- High leverage amplifies and propagates negative shocks more strongly.
- The economy can switch endogenously between basins of attraction, generating debt supercycles.
- The wealth distribution is central because it determines precautionary saving and the risk-free rate.
- A neural network approximates the nonlinear perceived law of motion for debt.
- The neural-network PLM is much more accurate than a linear PLM.
- The model can be evaluated with a likelihood function using inference with diffusions.
- The central lesson is that financial frictions and wealth heterogeneity jointly create nonlinear macro-financial dynamics.

7 Conclusion

7.1 Core conclusion

Fernández-Villaverde, Hurtado, and Nuño show that a heterogeneous-agent model with financial frictions can generate endogenous aggregate risk and regime-like macro-financial dynamics. The model moves between high- and low-leverage regions even though the deterministic steady state is unique.

7.2 Economic intuition

The intuition is a feedback loop. When leverage is high, experts are fragile because small capital losses strongly reduce net worth. Households understand that high leverage makes future wages and interest rates riskier, so they save more in bonds. This precautionary saving lowers the risk-free rate and sustains expert leverage. Thus the high-leverage region is internally consistent, as is the low-leverage region.

7.3 Takeaway

The paper's contribution is both economic and computational. Economically, it shows that the wealth distribution is a state variable for financial instability. Computationally, it shows how neural networks can solve nonlinear heterogeneous-agent models with aggregate shocks by approximating a flexible perceived law of motion.